



CODECHEF

The CodeChef.com Challenge, your monthly dose of coding puzzles from India's biggest coding contest, is now in print!

H

ello and welcome to the *CodeChef Puzzle Challenge* sponsored by *LINUX For You* and

Directif. Solve this month's puzzle, and get a chance to be one of the three lucky people to win a cash prize of Rs 1,000 each!

The April edition's puzzle:

Consider two stations A and B, located 100 km apart. There is a station C, situated between A and B. Now trains from station A and B start moving towards station C at different speeds. The train starting from A moves at a speed of 40 km/h while the train from B moves at a speed of 60 km/h. These trains have to stop at station C. A bird sets out from station B and flies at a speed of 80 km/h. When it reaches the train approaching from station A, it switches directions, going back the way it came. Then when it reaches the train coming from station B, it turns back again in the opposite direction. This goes on till both the trains reach station C. What is the total distance that the bird will have travelled by the time the two trains reach station C?

The solution:

Since the trains are travelling in opposite directions, the relative speed of the two trains = $40 + 60 = 100$ km/h .. (1)
The distance to be travelled by the

two trains = distance between two stations A and B = 100 km .. (2)

The time taken to travel 100 km = (Eqn 2)/(Eqn 1) = $100/100 = 1$ hr ..(3)

Speed at which the bird flies = 80 km/hr .. (4)

So distance travelled by the bird in 1 hr = (Eqn 4) * (Eqn 3) = $80 * 1 = 80$ km

And the winners this month are:

- Megha Gupta
- Ravikiran Gunale
- Ravat

Here's Your CodeChef Puzzle of the Month.

A man is standing on the banks of a river. He sees a temple on the opposite bank. He wants to visit the temple. He decides to row a boat across the river to the other bank. He can row at a speed of 4m/sec and the water in the river is flowing at 3m/sec. The man walks at a speed of 2m/sec. He starts rowing the boat straight across the river. Given that the width of the river is 800 metres, what is the time taken by the man to reach the temple?

Looking for a little extra cash?

Solve this month's *CodeChef Challenge* puzzle and send in your answer to codechef@efyindia.com or lly@codechef.com by May 15, 2012. Three lucky winners will get some awesome prizes both in cash and other merchandise! You can also participate in the contest by visiting the online space for the article at <https://www.facebook.com/LinuxForYou>



About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to CodeChef.com for the May Challenge that takes place from the 1st to the 11th of the month to win cash prizes up to Rs 20,000. Also keep visiting CodeChef and participate in the multiple programming contests that take place on the website throughout the month!